

CHAPTER 2.3 - GRID-FORMING INVERTERS: THE TRANSISTOR THAT LEARNED TO SING

On April 28, 2025, just past noon, Spain was doing exactly what energy optimists had been promising it would do. Solar panels were producing over half of the country's electricity. The sun was shining. The future was working.

And then it stopped. First, a solar farm in the south tripped offline. The voltage on the transmission network began to oscillate. Two large conventional generators that had been contracted to provide spinning reserve – the safety net the grid depends on when things go wrong – were not running. Nobody noticed in time. Within a minute, hundreds of solar plants disconnected in rapid succession, their inverters doing precisely what regulations required them to do: shutting down the moment voltage swung outside the permitted band. Within ninety seconds, Spain and Portugal went dark. Fifty million people without power. Trains stopped in tunnels. Traffic lights died. Hospitals switched to backup generators. The [largest blackout in Western European history](#).

In the aftermath, commentators lined up to blame solar energy. The sun, they said, was unreliable. Renewables had made the grid fragile. Some called for a return to coal and nuclear baseload. The [ENTSO-E final report](#) told a different story. The blackout was not caused by solar power. It was caused by [generators that should have been running but weren't](#) and by regulations that forced solar inverters to disconnect rather than help. My own additional conclusion was that the cause was an electrical system still designed around the assumption that stability comes from the top – from big, spinning machines – rather than from the millions of devices at the edges that could help to keep the system stable, and that can make sure most of the grid simply keeps running if one node trips. I thought of holons, but more about that later.

First let's talk about these new devices at the edges that can make the system distributed and stable.

Solar panels produce direct current. The grid runs on alternating current. The inverter is the translator between these two worlds, the connective tissue between every solar panel and the grid it feeds. In Spain, that translator had been shackled to rules written for an era of coal and steam. The inverters were *grid-following*: designed to ride a wave that someone else was making. When the wave broke, they had no choice but to let go. Solar had been made to follow a system that was never designed for it, instead of leading us into a future designed around what solar actually is: abundant, distributed, electronic. And it was all because we used the translator wrong.

This chapter is the story of that translator. Together with the battery from the previous chapter it can work miracles. It can stabilise voltage, set frequency, and respond to disturbances in milliseconds. Anyone with a solar system, a battery and an electric car can run a microgrid with no instructions from above. The inverter might be the most consequential, least celebrated machine in the whole energy transition.

But to understand why inverters are such a huge step forward, we need to start with the machine that has ruled the grid for a century and a half, and why, at the edges, its reign is ending.

One ring to rule them all

In 1831, Michael Faraday wrapped two coils of wire around opposite sides of an iron ring and discovered that a changing current in one coil induced a current in the other. He had built the world's first transformer, though it would take another half-century before anyone figured out what to do with it.

Once they found out, it enabled them to use Faraday's ring to win the War of the Currents – the bitter, occasionally deranged 1880s battle between Thomas Edison's direct current and Nikola Tesla and George Westinghouse's alternating current.

The transformer is in essence two sets of copper coils wound around a shared iron core. No moving parts. No fuel. No software. Just copper and iron enthusiastically obeying the laws of electromagnetism as if they had a choice in the matter. Feed alternating current into one coil and the oscillating magnetic field it creates in the iron core induces a current in the other coil. So far it seems rather useless, but here it comes: if you give the input side more or less windings than the output side, a transformer can change the voltage on the output side. More turns on the output side, voltage goes up. Fewer turns, voltage goes down. The mathematics is a first-year physics problem. The execution, at scale, is art.

The efficiency is astonishing. A large power transformer – the kind that sits in a high-voltage substation, humming quietly behind a chain-link fence – converts **99.5 to 99.7 per cent** of the energy fed into it. Distribution transformers, the smaller ones on power poles and in street cabinets, manage **98.9 to 99.5 per cent**. That kind of performance is brutally hard to achieve with active electronics and flatly impossible for any machine that burns fuel.

Edison's DC had no device that could perform the trick of changing the voltage. Everything worked at the same relatively low and safe voltage. This had the disadvantage that electricity could not travel more than a couple of kilometres before most energy was lost due to resistance in the wires. Thicker wires helped but the required thickness quickly became impractical and prohibitively expensive.

Now voltage comes in. Due to laws that describe the behavior of electricity flows, resistance in a wire is proportional to the current that flows through it, but independent of voltage. Bottom-line: if you increase the voltage, you increase the amount of energy you transport without needing thicker wires.

So alternating current can use a transformer to create very high voltages, make the electricity travel far with low losses through a relatively thin wire, then use another transformer to bring the voltage down again, et voilla your dishwasher is working.

During the trip this high voltage was of course dangerous, and Edison, in a last act of desperation, publicly electrocuted animals to prove this, including, in 1903, a circus elephant named Topsy.

But Edison's cruel theatrics couldn't change the physics and Tesla's AC won. The transformer electrified the world. And for the next 130 years, Faraday's rings sat at the heart of every grid on earth. Millions of them, connected by wires, stepping voltage up and down at every level of the system. One ring at the power station. Another at the substation. Another on the pole outside your house. The greatest machine ever built is, fundamentally, a mesh of Faraday's rings.

But there is a problem. Transformers are heavy. A 10 MVA substation transformer – the kind that serves a few thousand homes – weighs as much as a fully loaded shipping container. It arrives on a flatbed truck with a police escort. It gets bolted to a concrete pad and stays there for forty to sixty years. This is not a technology poised for disruption. This is a technology that has found its resting place.

The reason is physics. Specifically, 50 Hz physics. A transformer's iron core has to be large enough to carry the magnetic flux produced by the alternating current passing through it. At 50 cycles per second – 60 in the always exceptional United States – that requires a lot of iron. The magnetic flux density of grain-oriented electrical steel maxes out at about 1.5 to 1.8 tesla. You cannot cheat this. You can use better steel, you can optimise the geometry, but the fundamental relationship between frequency, flux, and core volume has not changed since Faraday did his first experiment.

Worse, the heaviness gets disproportionately worse as the device shrinks. A 10 MW transformer comes in at about 1.5 kilograms per kilowatt and is 99.5 per cent efficient. A 10-watt transformer, the kind that used to live in your laptop charger, comes in at about 30 kilograms per kilowatt and is around 65 per cent efficient. Your laptop charger used to be a warm brick because [50 Hz physics demanded that much iron and copper](#).

Then there is the second problem, which finishes the transformer's case. A transformer can only adjust voltage if it receives alternating current, and solar panels, batteries, computers, EVs and frankly most modern electrical devices run on direct current. Before transistors, if you actually had to convert DC to AC – say, to connect a battery bank to an AC grid – you used a [motor-generator set](#): an electric motor physically spinning a dynamo. A motor. Driving a generator. To convert one form of electricity into another. These contraptions were common in industrial settings and [submarine power systems](#) well into the 1960s. They vibrated, they wore out, they leaked oil, and they added five to ten per cent losses on top of whatever you were already losing. Connecting a bona fide electric motor to your solar panel just to create the AC a transformer needs is, when you think about it, not a winning solution.

And the limitations do not end there. A transformer cannot adjust frequency either. It cannot manage power factor. It cannot respond to grid conditions. It cannot be updated. The capabilities it has are the capabilities it was wound with, engraved on its nameplate the day it left the factory, unchanging until the day it is scrapped. For a century, this did not matter. The grid ran on alternating current from large spinning generators, and the one trick the transformer was capable of was the only trick anyone needed.

But we are leaving that grid behind. Solar panels produce DC. Batteries store DC. Electric vehicles run on DC. Computers run on DC. Heat pumps and modern appliances internally

rectify to DC. The world is going direct, the transformer can only speak alternating, and we need far more tricks than just voltage conversion. Faraday gave us a ring, and we wound a civilisation around it. But at the edges of the grid, where DC rules and the future is being assembled one rooftop at a time, a contender has shown up. One that does the transformer's job at a fraction of the weight, with a whole arsenal of other tricks up its sleeve.

The transistor that learned to sing

You might have heard about the transistor. It's the device that brought you computers, smartphones and the internet. It's basically a tiny switch that can be flicked on and off using tiny power flows. That's how it encodes zeros and ones.

What you might not know is that transistors are also used in power electronics where they switch large power flows. I became interested in such power transistors when I was designing high-quality class-D amplifiers for my self-built active loudspeakers. I am what they call an audiophile, which means I pay way too much attention to how music is reproduced. Class-D amplifiers were, at the time, the new up and coming new generation amplifiers and I was amazed at how well you can make a transistor sing. Literally. It was only later, when I started reading about grid-forming inverters, that I realised they use exactly the same mechanism as a class-D amplifier.

To convert DC to AC with a transistor, you take a steady DC voltage – say, 50 volts from a string of solar panels – and you flick it on and off thousands of times per second, generating pulses in a carefully shaped pattern. That pattern of pulses can then generate an AC sine wave. Wider pulses push the voltage up higher, narrower pulses bring the voltage down. A small inductor and capacitor smooth the wave further and you end up with clean, grid-quality AC. This technique is called pulse-width modulation.

Unlike a transformer, which must be physically rewound to change its voltage ratio, the inverter can change voltage, frequency and waveform in microseconds. All by adjusting the timing of those on-off pulses. All in firmware. And the device that does the actual switching is tiny compared to a traditional transformer. A [single power transistor module](#) – small enough to hold in one hand, weighing less than a kilogram – can switch over a megawatt. A transformer handling the same power? Over a hundred kilograms of copper wound around a half-tonne of iron core. A ratio of roughly two thousand to one. Imagine replacing a grand piano with a harmonica that sounds grander.

And the even bigger advantage is the repertoire. A transformer does one thing: voltage adjustment, at a ratio engraved on its nameplate. An inverter does all of the following, simultaneously, in milliseconds:

- *Voltage adjustment* – not a fixed ratio but a continuously adjustable range, changed on the fly.
- *DC to AC conversion* – the fundamental trick solar panels and batteries need, the one that used to require a spinning motor-generator set.

- *Frequency setting* – the inverter does not passively inherit the grid's 50 Hz. It can actively *set* it, pushing back against disturbances.
- *Power factor correction* – nudging the timing between voltage and current into alignment, reducing waste.
- *Waveform shaping* – producing whatever waveform the grid needs at that moment, not just a textbook sine.

When you have a solar system, all of this happens in a device the size of a shoebox bolted to your wall. None of that was possible in the era of the transformer. And because the inverter's behaviour is determined by software, it has something the transformer never will: a brain. A brain that can be upgraded over the internet. A software update can teach an old inverter new tricks. Which brings us to the trick that matters most for the holarthic bottom-up grid that this part of the book is about.

Grid-following vs grid-forming

There are two ways an inverter can connect to the grid. One is called grid-following. The other is called grid-forming.

A grid-following inverter listens. It measures the grid's voltage and frequency and synchronises to it. It's operating in lock step with the big boys that are allowed to think for themselves. Traditionally the coal plants, a gas turbines and big hydro generators. If the wave wobbles, the grid-following inverter wobbles too. If the wave breaks – if voltage swings outside the permitted band, or frequency strays too far from 50 Hz – the inverter does what its firmware and the grid code require. It disconnects. By design.

This was a perfectly sensible thing to require in 1990, when there were maybe a few hundred grid-connected inverters in the country and the grid's stability was guaranteed by a few dozen large rotating machines spinning in synchrony. The inverter was a guest. If something went wrong, the polite thing to do was leave the room.

By April 2025, Spain had tens of millions of grid-connected inverters. Solar covered more than half of national electricity production at noon on the day the grid went down. The polite guests were now the household. And their politeness – following the regulatory requirement to disconnect at the first sign of trouble – was the mechanism that turned a local oscillation into a continental blackout. Two large generators that should have been running for spinning reserve weren't. A voltage oscillation began. Hundreds of solar plants, each obeying the rules written for an earlier grid, dropped offline in succession. Each disconnection made the imbalance worse. The cascade took 90 seconds and then Spain went black.

A grid-forming inverter does not just listen to grid, it has a conversation with it. So there is listening involved – to the grid frequency mainly – but also talking. A grid forming inverters generates and maintains its own voltage and frequency using its internal clock and internal computer. If the bigger grid wobbles, a grid-forming inverter doesn't disconnect but gently the larger grid towards balance again. If the grid around a grid-forming inverter falls and nothing can be done anymore, a grid-forming inverter disconnects from the bigger grid and

keeps its own microgrid up and running. As long as the local power supply and batteries are enough, everything inside the microgrid is protected. Then, as soon as the sensors at its edges see that the grid has recuperated from its fall, the grid-forming inverter clasps hands with the larger grid again.

Now imagine every home and every neighbourhood being managed by such an inverter. Each contributing stabilization of voltage, frequency, reactive power and harmonic filtering to the wires it shares with its neighbours. Disturbances get absorbed locally, in milliseconds, by the nearest device. With such a system, the Iberian blackout would not just have been avoided, it would have been impossible. When something fails in such a grid, the failure does not cascade. It inconveniences whoever is directly connected to the failing node and not under the protection of a local microgrid, but nobody else is impacted.

So a grid-forming inverter provides the following capabilities:

- *Local voltage and frequency stabilisation* – the grid's most precious commodity, previously generated only by spinning iron in distant power stations, now available everywhere there is solar plus battery.
- *Black-start* – restarting a collapsed grid currently requires hours of manually sequenced re-energisation and is one of the most precarious operations in engineering. With grid-forming inverters, it becomes automatic. Each island energises itself from its own DC sources and automatically synchronises when it reconnects once the grid is up again.
- *Islanding* – when something upstream fails, the local section continues operating on its own power. Hospitals, neighbourhoods and homes become electrically resilient by default rather than by emergency procedure.
- *Reactive power and harmonic filtering* – the housekeeping functions that determine whether AC arrives clean. In a microgrid your expensive appliances can reliably get high quality electricity.

You might wonder how this dialogue between the grid-forming inverter and the larger grid works. How can this be done without a central conductor?

The answer is that they use the same mechanism the grid has always used to synchronise its large generators. Which incidently is the same mechanism Christiaan Huygens observed from his sickbed in 1665 when he noticed that two pendulum clocks hung from the same wooden beam would, after a few hours, swing in lockstep. It's called coupled oscillation. The internet is full of great video demonstrations.

Oscillators that are coupled because they share a medium pull each other into step. On today's grid, the medium is the alternating-current waveform itself, and the synchronisation mechanism is called droop control: as a generator gets loaded, it slows a fraction; as it gets unloaded, it speeds up a fraction; over hundreds of large generators connected to the same grid, the speeds converge. Grid-forming inverters do the same thing in firmware. What works for hundreds of large generators works equally well for millions of small inverters, because the physics does not care how big the oscillator is, only that it

shares the wave. The grid does not become more complicated. It just acquires more nodes. [Coupled-oscillator synchronisation has been confirmed in dozens of follow-up experiments since Huygens](#); the question is not whether it works but how to make operators certify it.

Unfortunately, almost every solar inverter operating in the world today is grid-following. The hardware of grid-following inverters is largely identical to grid-forming ones. The difference is firmware and certification. [IEA-PVPS Task 13](#) documented that the same power-electronics hardware that runs a grid-following algorithm can, in many cases, be upgraded to run a grid-forming one. New inverters can could ship with grid-forming capability at negligible additional hardware cost once certification became routine. The premium, [estimated at 25 to 50 per cent in 2022](#), is [converging toward zero](#).

Grid codes have been slow to acknowledge this new reality, and in my humble opinion that is what caused the Iberian blackout. And even now, most jurisdictions still require inverters to disconnect during disturbances – the exact behaviour that turned a local oscillation into a continental cascade. Fortunately, after the blackout of April 2025, this has begun to change. Grid codes in [Australia, the UK, and Germany](#) are moving toward requiring grid-forming capability for new installations.

Bottom-line: reason we don't use more grid-forming inverters today is institutional. The technology to prevent a repeat of Iberia exists, is commercially available, and is converging on free once grid operators accept and certify the new reality.

The incredibly shrinking inverter

Working on the energy transition is marvellous partly because you get to watch almost magical technological progress from so many directions. It might be too complex to easily explain to your uncle at birthday parties but it's highly motivating.

One recurring theme in this book has been how much less raw materials we need once we turn our back on fossil fuels. Solar panels and wind turbines were the first example. Batteries the second. But the inverter beats them hands down in that department.

My roof has eighty solar panels and a 30-kilowatt inverter that weighs about the same as one or two of those panels. In 2000, a transformer-based inverter at that power level would have weighed roughly 150 kilograms. Today's residential workhorses come in around 20 kilograms. The trajectory is clear. I've put it in a table, along with efficiency.

Year	Product	kg/kW	Efficiency
2006	SMA Sunny Boy 5000TL	6.2	95.5%
2011	SMA Sunny Boy 5000TL-20	5.0	97.0%
2020	SMA Sunny Boy 5.0	3.5	97.0%
2024	Huawei SUN2000-5KTL-L1	2.4	98.4%

If I fit a curve to these developments, it becomes clear that residential inverters halve their weight roughly every fourteen years. Two reasons drive it. The first is the [wide-bandgap semiconductor revolution](#) – replacing the silicon in the power transistor with silicon carbide (SiC) and gallium nitride (GaN). These materials let the transistor switch faster and

waste less energy doing it. Good old silicon switches at 5 to 20 kilohertz. SiC manages 50 to 150 kilohertz. GaN pushes that to 100 kilohertz to over a megahertz: a hundred times faster than the silicon it replaces, with 70 per cent less energy lost per switch.

Why does switching speed matter? Because it determines the size of the biggest electrical components. The transistor itself is tiny. The inductor and capacitor next to the transistor are much heavier. Their duty is to smooth the pulses that come out of the transistor. As frequency goes up you need smaller inductors and capacitors to do that. A hundredfold increase in switching frequency yields, after some real-world friction with core losses, roughly a fifty to eightyfold reduction in passive component weight.

If I – as a speaker designer – want to understand it, the analogy of enormous bass woofers versus tiny tweeters is irresistible. You might know that the size of a speaker is determined by how much low notes it can produce. There are some tricks but roughly speaking, if a loudspeaker is smaller, it will also produce less deep bass notes. With power transistors that switch at a higher frequency, you are replacing woofers with tweeters.

But it's not just switching at higher frequency that makes the inverter shrink. It's also the increasing efficiency. An efficiency increase from 99% to 99.5% sounds dull, until you realize this halves the cooling you need, which in turn halves the heatsink and possibly eliminates the fan. Most of what makes an inverter heavy, even more than the filters, is the aluminum heatsink that gets rid of the waste heat from those components. Less waste heat, less aluminium, less enclosure, less weight and size.

Efficiency	Loss at 5 kW (watts)	Heatsink mass (kg)
95%	250	~3.4
98%	100	~1.35
99%	50	~0.68
99.5%	25	~0.34

When you go from 95 per cent to 99.5 per cent efficiency, the heatsink shrinks by a factor of ten. That is the difference between a component you bolt to a wall and a component you include on a circuit board. Inverter efficiency has been moving at a sober but relentless pace: from 88 per cent in 1988 to 95 per cent by 2000. In the laboratory, inverters achieved 99 per cent efficiency by 2017 and that is now also available for best in class commercial inverters for your home. There is no Carnot-style hard physical limit to how high efficiency could become and there 99.9% (with a heatsink mass of 70 grams) is possible.

The US Department of Energy targets 0.07 kilograms per kilowatt for entire automotive inverters and BMW and Honda are already approaching it in production vehicles.

An inverter you might know well is your laptop charger. Your laptop charger from 2010 – a warm rectangular slab the weight of a textbook, full of iron and copper at 50 Hz physics – has been replaced over the last decade by USB-C chargers with a tenth of the weight and twice the power. And it's happening for larger inverters too. My 30 kW 20kg inverter could well be replaced by an alternative weighting just 4 kg in 2050.

In essence the inverter is becoming ubiquitous. Every device will be able to stabilise its own frequency and voltage. And inverters will become so small that we will not even notice we have replaced Faraday's rings with a device that is infinitely more capable.

The incredibly shrinking cost

A solar inverter today costs around \$42 per kilowatt at the wholesale level. The raw materials in it cost about \$4.60. The other 89 per cent is manufacturing, testing, certification, firmware development, warranty reserves, distribution, and margin, and since these need no material at all they continue to fall in price quickly too as volumes increase.

If the curve continues at the rate of the last fifteen years, the projection is suspicious, even for someone who knows learning curves:

Year	\$/kW
2025	\$42
2030	\$19
2035	\$9
2040	\$4.30
2050	\$1.00

At a dollar per kilowatt, my 30 kW inverter would cost just \$30 in 2050 and the cost of certification and warranty would dominate the bill. The contribution to my electricity bill would be less than a hundredth of a cent per kWh. The cost of converting DC to AC, of providing grid stability, of enabling solar panels and batteries to participate in the grid, is converging on zero.

What we are watching is the semiconductor revolution arriving to electric power flows. The most-manufactured object in human history – the transistor – having spent the better part of a century learning to carry information, has now learned to carry power. Wind taught the industry that learning curves apply to energy as well as to electronics. Solar steepened that curve. Batteries, in the previous chapter, steepened it again. The inverter steepens it once more. It's a cascade of increasingly fast moving improvements taking over the old grid by storm.

The sky is the limit

The conventional wisdom among grid experts is that inverters are fine for low voltage applications but do not work at higher voltages because transistors cannot handle high voltages.

However, do you know what sector is also switching to inverters? Ultra high voltage direct current (HVDC). That's the technology that is used for the huge cables that connect enormous wind parks to shore. Cables that can span a continent. These cables use the highest voltages in the entire power system. Sometimes up to 1 million volts. And they use inverters.

The solution is a clever architecture called modular multilevel converters (MMCs) that was introduced by Siemens circa 2010. It basically consists of hierarchical stacks of relatively small inverter modules. Some MMCs already contain tens of thousands of low voltage inverter modules. More modules in parallel increase the total power. More levels of the stack increase the voltage. In a way it's a giant distributed computer for power flows.

We'll come back to this nested modular architecture very often in this part 2 of the book, often under the name of holarchy. For now it's simply interesting to note that many small modules make it possible for MMCs to profit from the learning curve in the much bigger market of consumer inverters. The learning curve makes conventional wisdom outdated once again.

What is also interesting, is that this became possible through a technology called photonics. You already use photonics because fibre optic cables transport most of the data on the Internet these days and it's also a major enabler for hyperscale datacentres. Photonics is essentially computing that uses light particles instead of electrons. Photonics is vital for MMCs because how else do you talk to an inverter module that is working at one million volts? With photonics it becomes easy because light provides near-perfect galvanic isolation.

That said, the standard mid voltage grid transformer isn't going anywhere soon. It's very hard to beat it's combination of ruggedness and galvanic isolation.

But apart from that basic plumbing in the middle of the grid, inverters are the future. They are the automatic choice when you connect the DC of batteries, solar panels and electric vehicles. As inverters keep getting cheaper and more powerful while transformers become more expensive, they will displace more and more transformers. And once we allow them to be grid-forming, they will make the grid distributed and inherently stable.

That's especially true once the last member of our team joins in the next chapter.